

System Test Report

Date 17 Sep 2026 **Code** main at 0706b4c (PR #5), plus docs-only commit c961e3f
Machine Windows 11 · NVIDIA RTX A4000 **Database** shared Neon PostgreSQL

All checks pass
10 of 10 run · 0 skipped

All four services were up and talking to each other. Every unit suite, all three end-to-end scripts, and a full sign-up → create lead → AI score run through the real backend API passed against the live models. The lead → meeting → deal → deal score chain was also traced through real records in the database.

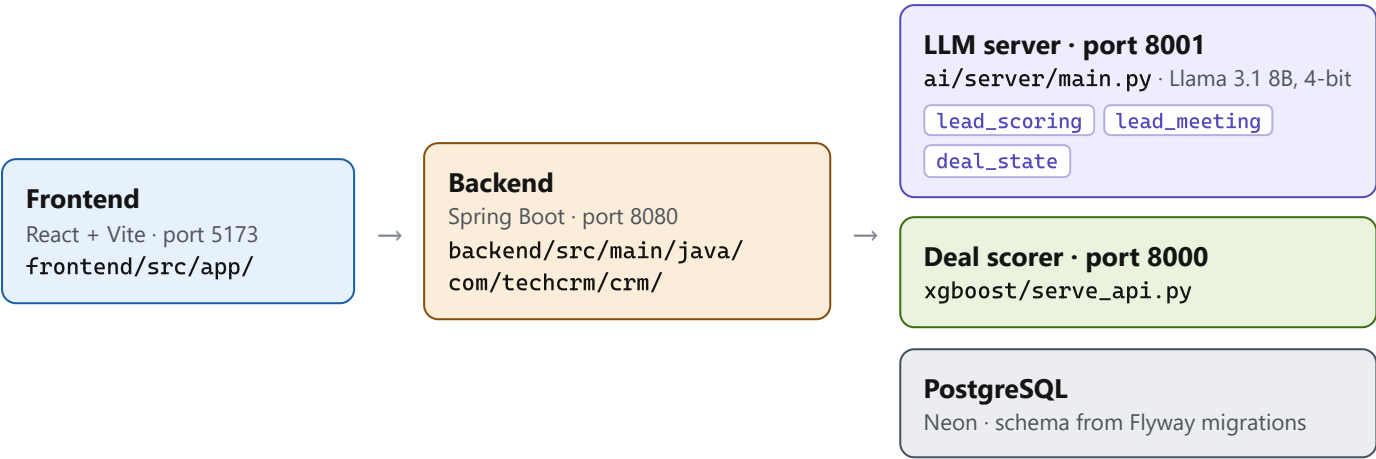
The full-stack test left one test organization (SystemTest d3ba3f15) with 3 leads in the shared database.

1 · Results at a glance

CHECK	WHERE	RESULT	DETAIL
Service health	/health on each service	4 / 4 up	Backend UP with database connected · LLM on CUDA with all 3 adapters · scorer model 1.0.0 · frontend HTTP 200
Backend unit tests	backend/ · mvnw test	80 passed	0 failures · BUILD SUCCESS
AI unit tests	ai/tests/ · pytest	143 passed	0 skipped. Server and bridge tests (74) in ai/venv; the 69 XGBoost contract tests in xgboost/.venv, which has pandas and xgboost
Frontend build	frontend/ · vite build	built	1,845 modules, no errors
Smoke test	ai/tools/smoke_test.py	3 / 3	Lead scoring 13.5 s · deal analysis 12.5 s · deal coach stream 10.8 s
System test · lead scoring	ai/tools/system_test.py	5 / 5 exact	Expected score and label matched for all five test leads
System test · deal chain	ai/tools/system_test.py	5 / 5 accepted	Deal-state adapter → XGBoost over 5 meetings: 46 → 53 → 78 → 67 → 80, 0 repairs
Live record trace	PostgreSQL, read-only	consistent	InnovateTech lead and deal checked step by step; see section 4
XGBoost unit tests	xgboost/tests/	32 passed	Run after installing pytest==9.1.1 (the pinned version) into xgboost/.venv
Full-stack API test	ai/tools/full_stack_test.py	3 / 3 exact	Signed up through /api/auth/signup, created 3 leads through /api/leads, all AI-scored: 100 Hot, 50 Warm, 0 Cold

2 · How the system fits together

The browser only talks to the backend. The backend is the only part that calls the two model services and the database.



In the flows below, backend files are under backend/src/main/java/com/techcrm/crm/ and frontend files under frontend/src/app/.

3 · Feature flows, step by step

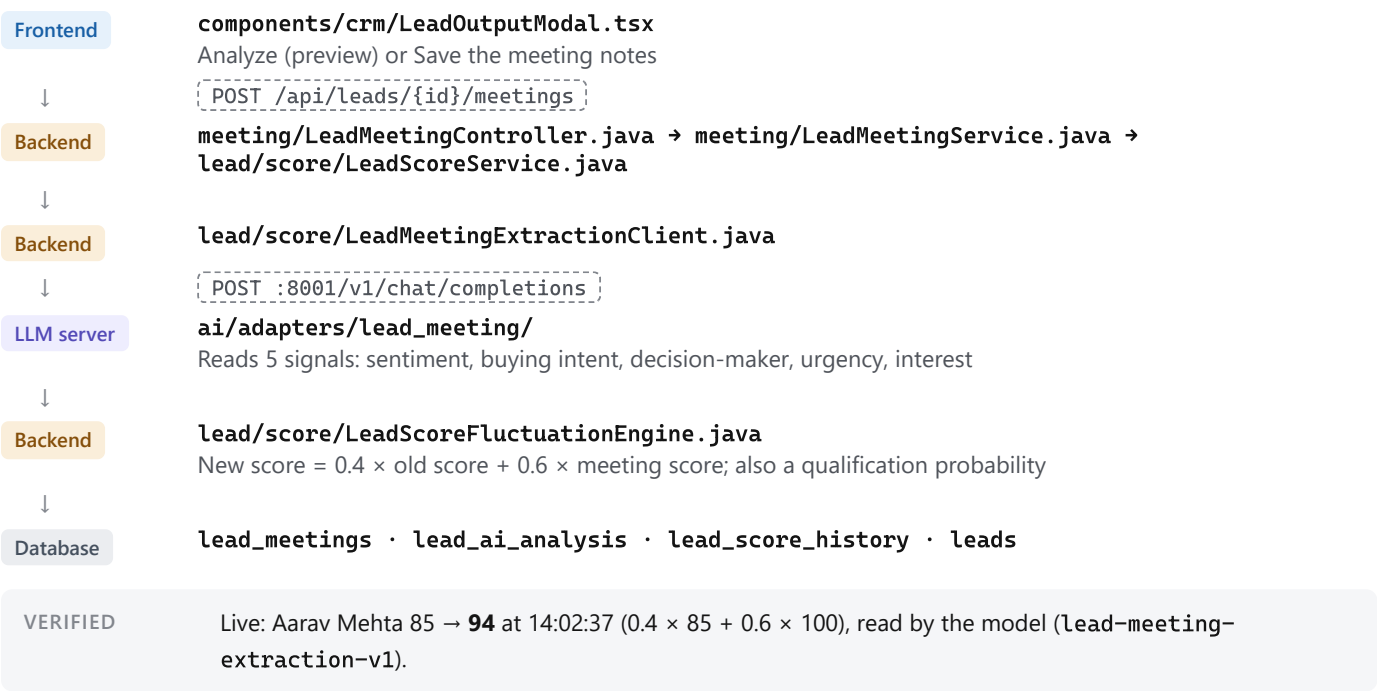
Each flow lists the files a request passes through, in order. Dashed labels are calls between services.

A · Add a lead and score it Lead management → Add lead

Frontend	screens/Wizards.tsx Lead form; sends the lead through lib/queries.ts POST /api/leads
Backend	lead/LeadController.java → lead/LeadService.java Builds the lead and asks for an AI score before saving
Backend	lead/AiScoringClient.java → ai/AiChatClient.java POST :8001/v1/chat/completions
LLM server	ai/server/main.py → ai/adapters/lead_scoring/prompt_format.py + weights/ : returns score 0–100 and Hot / Warm / Cold
Database	leads AI score, label, reason and status saved
VERIFIED	System test 5 / 5 exact. Live: Aarav Mehta scored 85 at 14:02:27.

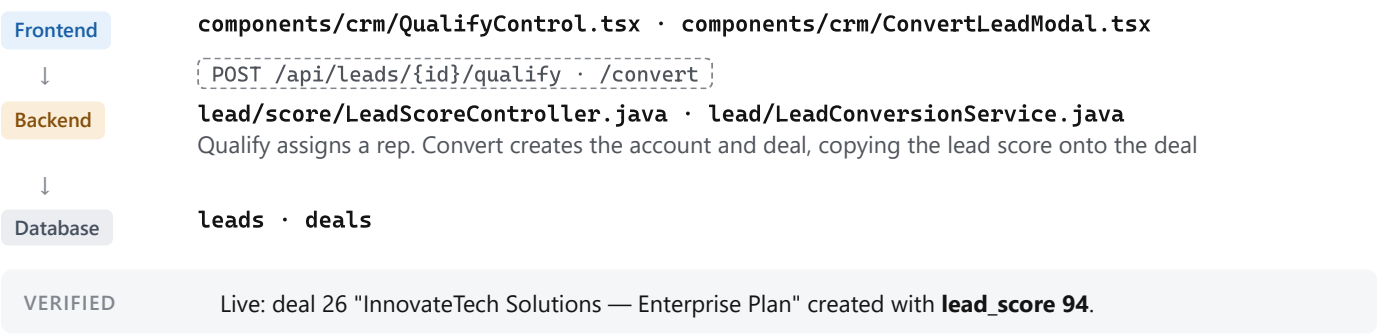
B · Record a lead meeting

Lead management → lead → meeting



C · Qualify and convert a lead to a deal

Lead list → Qualify, Convert



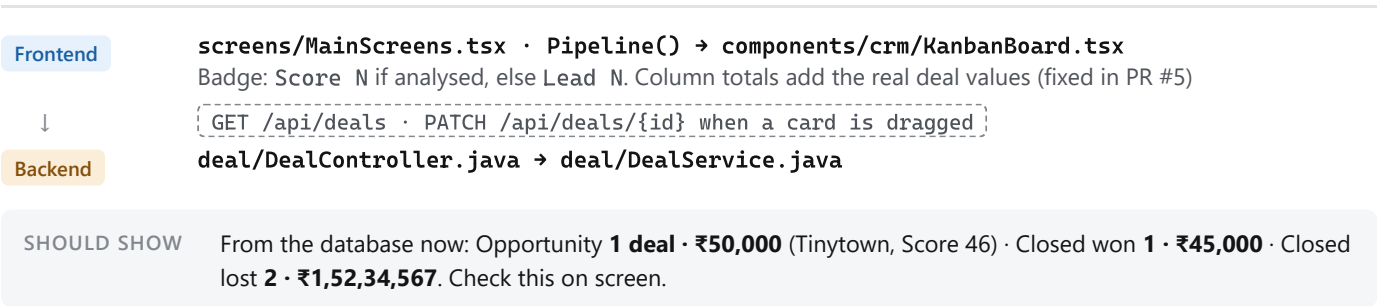
D · Deal meeting output → deal score

Sales pipeline → Open workspace → meeting output



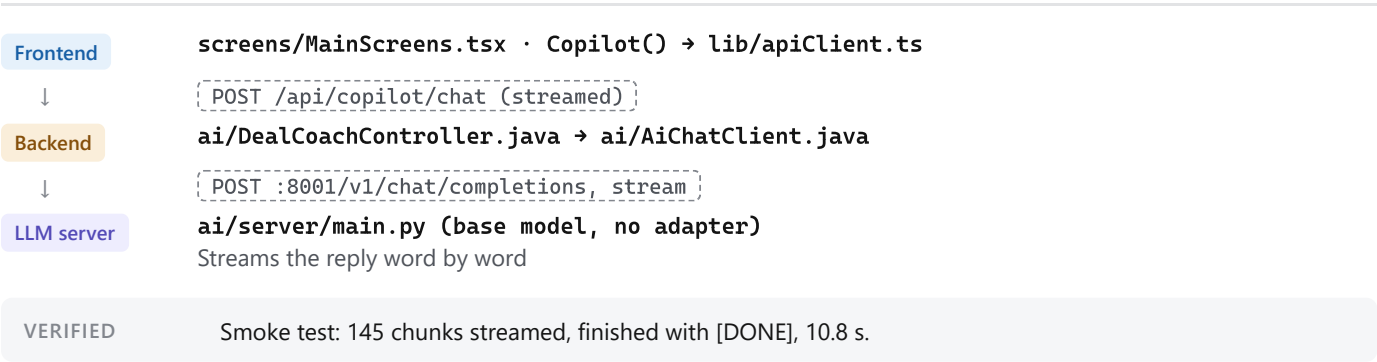
E · Sales pipeline board

Sales pipeline



F · Deal coach chat

Deal coach



4 · One real customer, end to end

Aarav Mehta / InnovateTech Solutions Pvt. Ltd., from the database (read-only). Times are IST on 17 Sep 2026.

TIME	STEP	HANDLED BY	RESULT
14:02:27	Lead created	lead_scoring	Score 85, HOT
14:02:37	Qualification meeting	lead_meeting	Decision-maker Present, intent High, urgency High → score 94 , qualification probability 100
–	Qualified and converted	LeadConversionService	Deal 26, value ₹2,34,567, lead score 94
14:22:21	Deal meeting output	deal_state	SUCCEEDED in 18.3 s, 14 parameters extracted, 0 errors
14:22:26	Deal scored	XGBoost	Deal score 47.85 , band LOW, win probability 47.6 %, risk HIGH
–	Current stage	Pipeline board	Closed lost

5 · System test: deal chain detail

Five meetings for one simulated deal, sent through the deal-state adapter and then the XGBoost scorer (ai/tools/system_test.py).

#	MEETING	DEAL SCORE	BAND	FIELDS CHANGED	REPAIRS	TIME
1	Opening call	46.3	LOW	4	0	17.4 s
2	Demo, CFO joins	52.9	MEDIUM	6	0	17.2 s
3	Budget approved	77.6	HIGH	5	0	17.2 s
4	Objections raised	67.2	MEDIUM	7	0	16.8 s
5	Verbal agreement	80.1	HIGH	6	0	17.7 s

The score rises overall (+33.8) and dips at the objections meeting, as expected.

6 · Folder map

CRM/	
frontend/src/app/	React app (port 5173)
screens/	Wizards.tsx, MainScreens.tsx
components/crm/	LeadOutputModal, ConvertLeadModal, DealWorkspace, KanbanBoard ...
lib/	queries.ts (REST calls), apiClient.ts (streaming)
backend/src/main/java/com/techcrm/crm/	Spring Boot API (port 8080)
lead/ lead/score/	lead scoring, meetings engine, conversion
meeting/	lead meeting endpoints
dealflow/ deal/	deal meeting pipeline, deal scoring client
ai/	AiChatClient, DealCoachController
ai/	LLM server (port 8001)
server/main.py	serves all three adapters
base-model/	Llama-3.1-8B-Instruct (not in git)
adapters/lead_scoring/	prompt_format.py train.py data/ eval/ weights/
adapters/lead_meeting/	prompt_format.py train.py data/ eval/ weights/
adapters/deal_state/	prompt_format.py train.py data/ eval/ weights/ README.md
tests/	test_server, test_bridge, test_contract
tools/	smoke_test, system_test, full_stack_test, inference
xgboost/	deal scorer (port 8000)
serve_api.py deal_score_service.py deal_score_pipeline.py	
outputs/xgboost_deal_score.pkl	
docs/	DATABASE-SETUP.md, ATTRIBUTIONS.md, reports

7 · Open items

Before the next release

- **Check the pipeline board on screen** against section 3E (the fix from PR #5 hasn't been viewed in a browser).
- **Push backup-restore-docs** and merge it: restores DATABASE-SETUP.md and ATTRIBUTIONS.md.
- **Test data in the shared database:** organization SystemTest_d3ba3f15 (user systest.d3ba3f15@example.test) with leads 298–300. It's scoped to its own organization, so other users don't see it; delete it if you want the database clean.

Clean-up

- The contract tests only run in a venv with pandas; use xgboost/.venv, or install ai/requirements-train.txt into ai/venv.
- Old paths and port 8002 remain in some backend comments, application.yml and the smoke-test start hint.

8 · How to run these checks again

```
cd D:\CRM\ai
venv\Scripts\python server\main.py                # terminal 1: LLM server
cd D:\CRM\xgboost
.venv\Scripts\python -m uvicorn serve_api:app --port 8000  # terminal 2: deal scorer

cd D:\CRM\ai                # terminal 3: tests
venv\Scripts\python -m pytest tests -q
venv\Scripts\python tools\smoke_test.py
venv\Scripts\python tools\system_test.py
venv\Scripts\python tools\full_stack_test.py        # needs backend :8080; creates a test org
..\xgboost\.venv\Scripts\python -m pytest tests\test_contract.py -q
cd D:\CRM\xgboost
.venv\Scripts\python -m pytest tests -q
cd D:\CRM\backend
.\mvnw.cmd test
```

Generated 17 Sep 2026. Test output from this machine; database figures from read-only queries against the shared Neon instance.